DANNY LIU

MECHANICAL ENGINEER

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Mechanical Engineer with experience in thermal systems, heat transfer, and airflow analysis, with a focus on cooling technologies and energy-efficient design. Skilled in CFD simulation, CAD modeling, and experimental testing to improve thermal performance and reliability.

AREA OF EXPERTISE

Battery Thermal Management	Simulation & Modeling	Materials Engineering
Heat Transfer Engineering	CAD & Mechanical Design	Experimental Testing

PROFESSIONAL EXPERIENCE

University of Wisconsin-Milwaukee— *Graduate Assistant* **Aug 2024 – Present**

- Lead and mentor students in computer-aided design (CAD) using SolidWorks and Creo, helping them develop practical engineering design skills.
- Conduct mechanical stress simulations of all-solid-state batteries using ANSYS Mechanical, supporting research in battery safety and performance.

Asia Vital Components Co., Ltd — *Fan & Thermal Systems* **Oct 2023 – June 2024**

- Designed high-efficiency fan blades and cooling systems using Creo Parametric to improve airflow, heat transfer, and energy efficiency in compact electronic and battery systems.
- Used ANSYS Fluent to run CFD simulations to analyze airflow, pressure distribution, and heat dissipation, helping optimize thermal performance and reduce aerodynamic noise.
- Worked with thermal engineers to develop cooling modules and verified design performance by comparing simulation results with prototype testing.

Industrial Technology Research Institute — *Surface Engineering for Heat Exchange* **May 2023 – Oct 2023**

- Developed nano-coatings and surface treatments to enhance heat exchanger and condenser performance, improving heat transfer efficiency in thermal and energy systems.
- Applied hydrophobic and hydrophilic surface engineering to improve condensation efficiency for dehumidification and cooling applications.
- Designed fin structures using Autodesk Inventor to improve heat dissipation and tested coating durability, adhesion, and thermal performance using laboratory instruments.

EDUCATION

University of Wisconsin–Milwaukee, Milwaukee, WI **Expected May 2026**

Master of Science in Mechanical Engineering & Industrial Engineering

- Coursework: Thermodynamics, Computational Fluid Dynamics (CFD), Six Sigma, Operations Research

Chung Yuan Christian University, Taoyuan, Taiwan **Sep 2020 - June 2024**

Bachelor & Master of Science in Mechanical Engineering (Accelerated Program)

- Research in battery thermal management and cooling technologies.
- Thesis on “Thermal performance optimization of lithium-ion batteries using misaligned multi-minichannels”.

ADDITIONAL INFORMATION

- **Languages:** English, Mandarin.
- **Technical Skills:** Ansys Fluent, Creo Parametric, SolidWorks, Autodesk Inventor, MATLAB.
- **Awards/Activities:** UWM Chancellor’s Graduate Student Award (2024), Outstanding Graduate Award at CYCU (2024)